Government Scheme Assistance Bot Project Research & Implementation Report

Team name: Vision X

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1. Scope of Work

Project Title

Government Scheme Assistance Bot

Objective

The Government Scheme Assistance Bot is an AI-powered chatbot designed to help citizens easily discover government schemes, determine eligibility, understand required documents, and receive updates regarding various government welfare programs.

The chatbot aims to reduce the difficulty citizens face when searching through government websites and lengthy documents.

Scope

The chatbot will provide:

- Government scheme discovery
- Eligibility checking
- Required document information
- Scheme updates
- Frequently Asked Questions (FAQs)
- Natural language interaction

2. Features

Core Features

1. Scheme Search

Users can ask:

"What schemes are available for farmers?"

"Government schemes for students"

The bot retrieves relevant schemes.

2. Eligibility Checker

Users provide details:

- Age
- Gender
- Occupation
- Income
- State

The bot determines eligible schemes.

3. Required Documents Information

The bot asks the required documents for the schemes like:

- Aadhaar Card
- PAN Card
- Income Certificate
- Caste Certificate
- Residence Certificate

depending on scheme requirements.

4. Scheme Updates

The chatbot informs users about:

- New schemes
- Updated eligibility criteria
- Application deadlines

5. FAQ Support

Examples:

- What is PM Kisan?
- Who can apply?
- How much benefit is provided?

6. Conversational Search

Users can ask naturally:

I am a farmer from Karnataka.
Which schemes can help me?

The bot understands context.

3. Libraries Used

1.Backend Development

Library	**Purpose**
Python	Main Programming Language
FastAPI	Backend API Development
Uvicorn	Server Hosting
Pydantic	Data Validation

2.AI & NLP

Library	**Purpose**
LangChain	RAG Pipeline
Sentence Transformers	Text Embedding
Transformers	NLP Models

3.Vector Database

Library	**Purpose**
ChromaDB	Store Embeddings

Library	Purpose
FAISS	Fast Similarity Search

4.Data Processing

Library	Purpose
Pandas	Data Handling
NumPy	Numerical Operations
PyPDF2	PDF Extraction
Unstructured	Document Parsing

5.Frontend

Library	Purpose
Streamlit	Chat Interface

4. Necessary Resources / Requirements

1.Dataset Requirements

Government scheme documents:

Examples:

- PM Kisan
- PMAY
- Ayushman Bharat
- Startup India
- Mudra Yojana
- National Scholarship Schemes

Document formats:

- PDF
- DOCX
- Government website content

2.AI Model Requirements

1.Embedding Model

Recommended:

all-MiniLM-L6-v2

Advantages:

- Fast
- Free
- Lightweight

2.LLM used

Gemini 2.5 Flash (via Gemini API)

5. Resources Used

1.Information Sources

Government Websites

- National Portal of India
- MyGov
- PM Kisan Portal
- Ministry Websites
- State Government Websites

Documents

- Scheme Guidelines
- Eligibility Documents
- Government Notifications
- FAQs

2.Knowledge Base

All documents will be converted into embeddings and stored in the vector database.

6. Steps Involved to Complete the Chatbot

Phase 1: Requirement Analysis

- Identify user requirements
- Identify supported schemes
- Define chatbot scope

Output:

Requirement Specification Document

Phase 2: Data Collection

Collect:

- Scheme documents
- FAQs
- Eligibility criteria
- Required documents

Output:

Raw Knowledge Base

Phase 3: Data Preprocessing

Tasks:

- Extract text from PDFs
- Remove noise
- Clean text
- Chunk documents

Output:

Clean Chunks

Phase 4: Embedding Generation

Convert text chunks into vectors.

Example:

Document Chunk
↓
Embedding Model

↓
Vector Representation

Output:

Vector Data

Phase 5: Vector Database Creation

Store vectors inside:

ChromaDB
or
FAISS

Output:

Searchable Knowledge Base

Phase 6: RAG Development

Develop Retrieval-Augmented Generation Pipeline.

Flow:

User Query
↓
Embedding
↓
Vector Search
↓
Relevant Documents
↓
LLM
↓
Answer

Phase 7: Chat Interface

Develop Streamlit chatbot.

Features:

- Chat window
- Query history
- User-friendly UI

Phase 8: Testing

Test:

- Eligibility Questions
- Scheme Search
- Updates
- FAQs

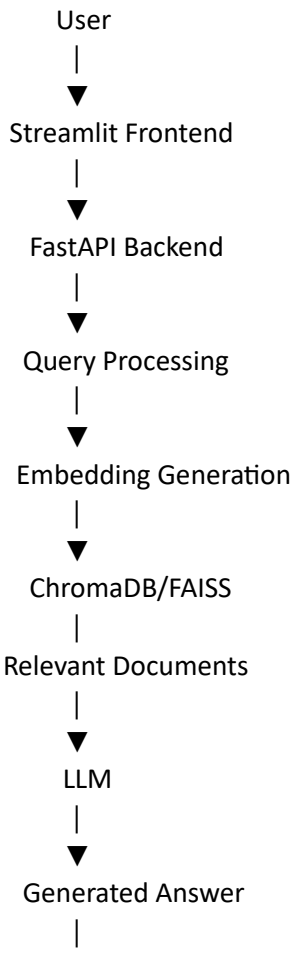
Phase 9: Deployment

Deploy:

- Localhost
- Cloud Server
- College Demonstration Environment

7. How the Chatbot is Implemented

1. System Architecture



▼

User

2. Workflow

Step 1

User enters question.

Example:

Which schemes are available for women entrepreneurs?

Step 2

Query converted into embedding.

Step 3

Vector DB searches most relevant chunks.

Step 4

Relevant scheme information retrieved.

Step 5

LLM receives:

- User query
- Retrieved documents

Step 6

LLM generates contextual answer.

Step 7

Answer displayed to user.

8. Concept Note – Logic Behind the Chatbot

1. Problem

Citizens often face difficulties:

- Searching government websites
- Understanding eligibility criteria
- Finding required documents
- Keeping track of scheme updates

The information exists but is scattered across multiple sources.

2. Proposed Solution

Create an AI-powered Government Scheme Assistant using the RAG (Retrieval-Augmented Generation) approach.

Instead of relying only on the LLM's knowledge, the chatbot first searches a trusted government knowledge base and then generates answers.

This reduces hallucinations and increases accuracy.

3. Core Logic

User Question

Query Embedding

Vector Search

Retrieve Relevant Schemes

Provide Context to LLM

Generate Accurate Answer

Return Response

4. Why RAG is Necessary

Without RAG:

LLM
↓
May Hallucinate
↓
Incorrect Government Information

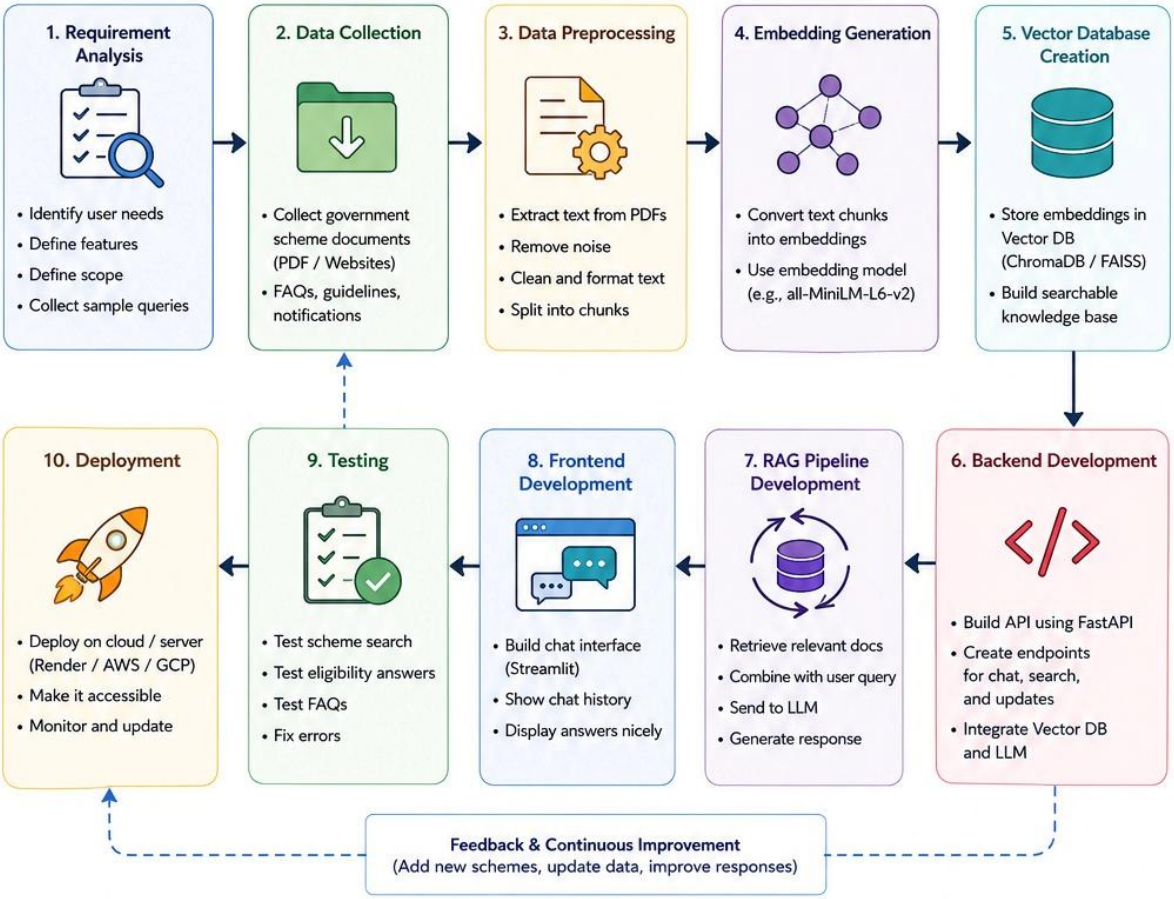
With RAG:

Government Documents
↓
Vector Database
↓
Retrieved Context
↓
LLM
↓
Accurate Answer

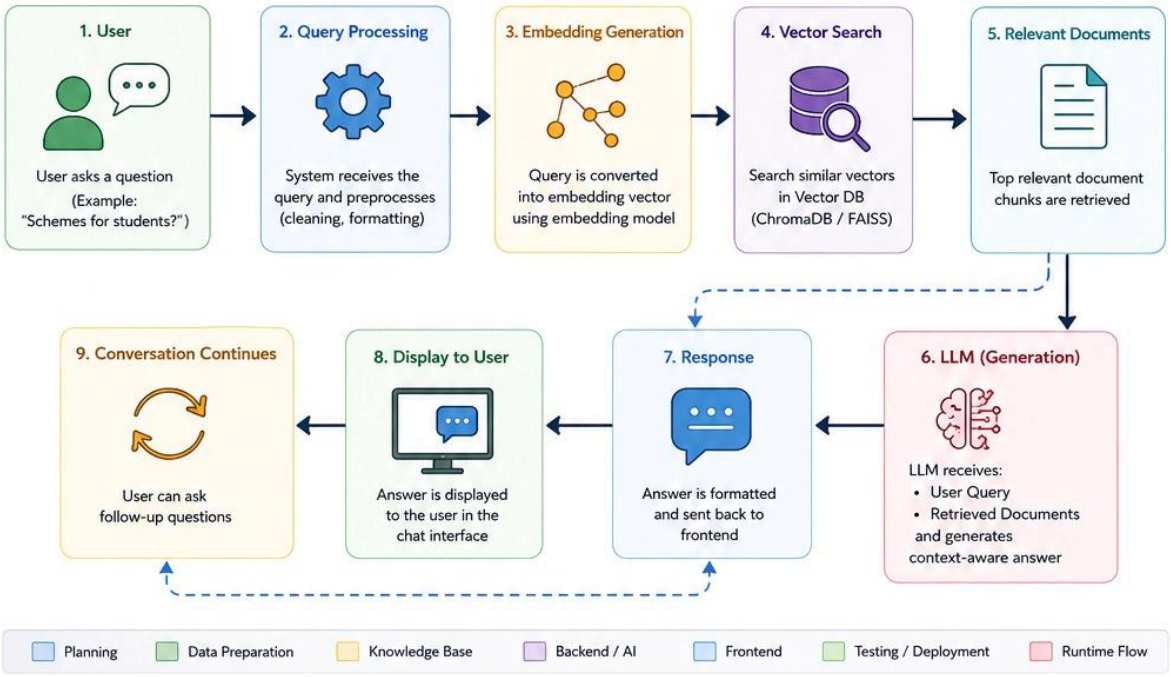
5. Recommended Technology Stack (Final)

Component	**Technology**
Programming Language	Python
Frontend	Streamlit
Backend	FastAPI
Embedding Model	all-MiniLM-L6-v2
Vector Database	ChromaDB
LLM	Gemini
Document Loader	PyPDF2
Framework	LangChain
Deployment	Local Server / Cloud

Flowchart – How We Will Build the Government Scheme Assistance Bot



Detailed Workflow (User Query to Answer)



THANK YOU